Temporal PhiID: Proposed Higher-Order Metrics for a Single-Signal Embedding

Overview

This document proposes a set of **new derived metrics** for **temporal PhiID** when PhiID is applied not to two separate systems, but to a **Takens delay embedding of a single time series**.

In standard PhiID, atoms are often grouped into metrics such as storage, transfer, erasure, and integrated information. These groupings are useful, but in the **temporal single-process case**, some of them become less natural because:

- “X” and “Y” are not different systems,
- “transfer” is not really inter-system transfer,
- the most important structure is often the **temporal geometry of information** across the embedding window.

The goal of this document is therefore to define **more adapted higher-order metrics** that summarize the 16 atoms in ways that are especially meaningful for:

- persistence,
- phase asymmetry,
- temporal integration,
- structural reformatting,
- broadcasting vs fragmentation,
- and multiscale temporal organization.

---

1. Temporal Embedding Reminder

We use a 4-point embedding of a single signal:

```
p1 = x(t)
p2 = x(t+ )
t1 = x(t+2 )
t2 = x(t+3 )
```

with timeline:

```
t     t+     t+2     t+3
p1      p2        t1          t2
```

We map this into PhiID as:

```
Sources (past): p1, p2
Targets (future): t1, t2
```

So the 16 atoms still exist, but they now describe how information **changes form across a temporal window of one process**.

---

2. The 16 Atoms

	→ r	→ x	→ y	→ s
r →	`rtr`	`rtx`	`rty`	`rts`
x →	`xtr`	`xtx`	`xty`	`xts`
y →	`ytr`	`ytx`	`yty`	`yts`
s →	`str`	`stx`	`sty`	`sts`

Where:

- `r` = redundant information
- `x` = information unique to `p1`
- `y` = information unique to `p2`
- `s` = synergistic information

---

3. Guiding Principle for New Metrics

In the temporal case, it is often more useful to ask:

- Does information **stay in the same form**, or change form?
- Does it remain **local**, become **shared**, or become **integrated**?
- Does the embedding show **temporal symmetry** or **directional imbalance**?
- Is the system mainly **preserving**, **assembling**, **broadcasting**, or **fragmenting** information?
- How do these structures evolve as a function of ?

The metrics below are designed to answer these questions directly.

---

4. Proposed Higher-Order Metrics

4.1 Persistence Hierarchy

Instead of one single "storage" metric, separate persistence into three levels.

4.1.1 Global Persistence

`Global_persistence = rtr`

Interpretation:
Information that remains redundant across the whole embedding window. This reflects **broad temporal persistence** or **global memory**.

High values suggest: - strong slow structure, - stable background state, - information that is present across all four positions.

4.1.2 Phase-Specific Persistence

`Phase_persistence = xtx + yty`

Interpretation:
Information that remains tied to one phase class across the window: - `xtx`: p1 → t1 - `yty`: p2 → t2

This reflects **lag-2 persistence within each phase channel**.

High values suggest: - skip-lag predictability, - oscillatory or alternating structure, - stable local phase identity.

4.1.3 Integrated Persistence

`Integrated_persistence = sts`

Interpretation:
Information that is synergistic in the past and remains synergistic in the future.

This reflects **persistent temporal integration**, i.e. information that can only be decoded by considering combinations of positions on both sides of the embedding.

High values suggest: - stable temporal wholes, - nonlocal temporal structure, - coherent integrated motifs.

4.1.4 Total Persistence

`Total_persistence = rtr + xtx + yty + sts`

Interpretation:
All information that preserves its structural type across the embedding.

This is the temporal analogue of **self-consistency** or **closure of information type**.

4.2 Structural Lability

```
Structural_lability =
rtx + rty + rts +
xtr + xty + xts +
ytr + ytx + yts +
str + stx + sty
```

Interpretation:
All atoms where information **changes type** from source to target.

This measures how much the system **reformats information** over the temporal window rather than simply preserving it.

High values suggest: - dynamic reorganization, - temporal transformation, - regime transitions, - structural instability or flexibility.

4.3 Closure vs Reformatting

Closure

```
Closure = rtr + xtx + yty + sts
```

Reformatting

```
Reformatting = Total_atoms - Closure
```

or explicitly:

```
Reformatting =
rtx + rty + rts +
xtr + xty + xts +
ytr + ytx + yts +
str + stx + sty
```

Interpretation:
This is a simple but powerful contrast:

- **Closure** = information stays in the same representational form
- **Reformatting** = information changes form across the window

This distinction is often more meaningful in temporal PhiID than the usual storage/transfer split.

4.4 Phase Exchange

Unsigned phase exchange

```
Phase_exchange = xty + ytx
```

Interpretation:
Information switches between the two phase channels: - `xty`: X-like past becomes Y-like future - `ytx`: Y-like past becomes X-like future

This measures **even/odd exchange**, or **cross-phase coupling**.

High values suggest: - oscillatory structure, - alternation, - parity coupling across the embedding.

Signed phase bias

```
Phase_bias = xty - ytx
```

Interpretation:
Whether the exchange is balanced or directionally biased.

- positive: stronger `p1 → t2` style mapping
- negative: stronger `p2 → t1` style mapping

Useful for detecting subtle temporal asymmetries.

4.5 Broadcast Index

```
Broadcast = xtr + ytr + str
```

Interpretation:
Information that ends up **redundant in the future**, regardless of whether it began as unique or synergistic.

This captures the tendency of the process to **spread information across future positions**.

Components: - `xtr`: local early info becomes shared - `ytr`: local mid info becomes shared - `str`: integrated info becomes shared

High values suggest: - convergence toward common future structure, - temporal homogenization, - stabilization into a shared state.

4.6 Fragmentation Index

```
Fragmentation = rtx + rty + stx + sty
```

Interpretation:
Information that begins as a more distributed structure and ends up localized to one phase channel.

Components: - `rtx`, `rty`: shared info breaks into one-sided persistence - `stx`, `sty`: synergistic info collapses into phase-specific information

High values suggest: - decomposition of global structure, - specialization, - breakdown of shared or integrated temporal organization.

4.7 Integration Formation

`Integration_formation = rts + xts + yts`

Interpretation:
Information that becomes synergistic in the future.

This measures the extent to which: - shared information, - phase-local information, - or both

are being assembled into **higher-order temporal patterns**.

High values suggest: - emergence, - temporal binding, - nonlinear joint predictability, - formation of integrated motifs.

4.8 Integration Deployment

`Integration_deployment = str + stx + sty`

Interpretation:
Synergistic information in the past that gets expressed in simpler forms in the future.

This is not necessarily "loss." It can reflect: - unpacking of a temporal whole, - projection of integrated structure into local channels, - stabilization of an emergent pattern.

High values suggest: - top-down constraint, - resolution of temporal integration, - expression of holistic structure into simpler observables.

4.9 Net Integration Balance

```
Net_integration_balance =
(rts + xts + yts) - (str + stx + sty)
```

Interpretation:
Whether the temporal window is, overall:

- **building integration** (positive),
- or **deploying / dissolving integration** (negative).

This is a very useful summary metric.

Positive values

The embedding tends to **assemble** information into temporally integrated patterns.

Negative values

The embedding tends to **unpack** integrated information into more local or shared forms.

Near zero

Balanced exchange between integration formation and integration deployment.

4.10 Temporal Asymmetry

Source-side asymmetry

```
Source_asymmetry =
(xtr + xtx + xty + xts) -
(ytr + ytx + yty + yts)
```

Interpretation:
Whether the earliest source position (`p1`) or the next source position (`p2`) contributes more strongly to the temporal information dynamics.

- positive: `p1` structurally dominates
- negative: `p2` structurally dominates

This can be useful for detecting: - asymmetry in temporal influence, - anticipation vs continuation, - edge effects in the embedding.

Target-side asymmetry

```
Target_asymmetry =
(rtx + xtx + ytx + stx) -
(rty + xty + yty + sty)
```

Interpretation:
Whether future information preferentially lands in the `t1`-like or `t2`-like phase channel.

This gives a measure of **future-phase imbalance**.

4.11 Shared-to-Local Decay

```
Shared_to_local_decay = rtx + rty
```

Interpretation:
Redundant information in the past that no longer remains shared in the future, but survives only in one phase channel.

This is a refined version of "erasure."

It specifically captures: - loss of broad temporal memory, - narrowing of a shared representation, - partial forgetting.

4.12 Synergy-to-Local Collapse

```
Synergy_to_local_collapse = stx + sty
```

Interpretation:
Integrated temporal structure that does not remain integrated, but instead collapses into one phase channel.

This can be interpreted as: - breakdown of holistic dynamics, - specialization of an integrated pattern, - local readout of a global temporal state.

4.13 Synergy Maintenance Ratio

```
Synergy_maintenance_ratio = sts / (rts + xts + yts + sts + str + stx + sty)
```

Interpretation:
Among all synergy-related dynamics, how much corresponds to **synergy remaining synergy**?

High values suggest: - stable integrated temporal motifs, - persistent joint structure.

Low values suggest: - transient or unstable synergy, - integration that is quickly created or dissolved.

4.14 Broadcast-to-Fragmentation Ratio

```
Broadcast_fragmentation_ratio =
(xtr + ytr + str) / (rtx + rty + stx + sty)
```

Interpretation:
Whether the temporal dynamics tend more toward:

- **spreading and sharing information** across future positions, or
- **breaking down distributed structure** into local channels.

This provides a compact measure of whether the system is becoming more globally coherent or more locally specialized.

---

5. Recommended Metric Families

A practical way to organize the atoms in temporal PhiID is into six families.

5.1 Persistence Family

`rtr, xtx, yty, sts`

Meaning:
Information preserves its structural type.

Suggested derived metrics: - `Global_persistence` - `Phase_persistence` - `Integrated_persistence` - `Total_persistence`

5.2 Exchange Family

`xty, ytx`

Meaning:
Information moves between phase channels.

Suggested derived metrics: - `Phase_exchange` - `Phase_bias`

5.3 Broadcast Family

`xtr, ytr, str`

Meaning:
Information becomes broadly shared in the future.

Suggested derived metrics: - `Broadcast` - `Broadcast_fragmentation_ratio`

5.4 Fragmentation Family

`rtx, rty, stx, sty`

Meaning:
Distributed structure becomes localized.

Suggested derived metrics: - `Fragmentation` - `Shared_to_local_decay` - `Synergy_to_local_collapse`

5.5 Integration Formation Family

`rts, xts, yts`

Meaning:
Information becomes synergistic.

Suggested derived metrics: - `Integration_formation` - contribution-specific versions: - `Redundancy_to_integration = rts` - `X_to_integration = xts` - `Y_to_integration = yts`

5.6 Integration Deployment Family

`str, stx, sty`

Meaning:
Synergy is resolved into simpler future structures.

Suggested derived metrics: - `Integration_deployment` - contribution-specific versions: - `Integration_to_shared = str` - `Integration_to_X = stx` - `Integration_to_Y = sty`

---

6. Minimal Core Set of New Metrics

If only a few new metrics are kept, the following set is probably the most useful.

Core set

1. Total Persistence

`rtr + xtx + yty + sts`

How much information keeps the same structural identity.

2. Structural Lability

`sum of all off-diagonal atoms`

How much information changes form across the window.

3. Phase Exchange

`xty + ytx`

How much information swaps phase channel.

4. Broadcast

`xtr + ytr + str`

How much information becomes broadly shared.

5. Fragmentation

`rtx + rty + stx + sty`

How much distributed structure collapses into local channels.

6. Integration Formation

`rts + xts + yts`

How much information becomes synergistically integrated.

7. Integration Deployment

`str + stx + sty`

How much synergistic information is unpacked or resolved.

8. Net Integration Balance

`(rts + xts + yts) - (str + stx + sty)`

Whether the temporal window is integration-building or integration-resolving.

---

7. Suggested Interpretation Table

Metric	Atom Combination	Temporal Meaning
Global_persistence	**rtr**	Broad memory shared across all four positions

Metric	Atom Combination	Temporal Meaning
Phase_persistence	`xtx + yty`	Persistence within each lag-2 phase channel
Integrated_persistence	`sts`	Stable temporal integration
Total_persistence	`rtr + xtx + yty + sts`	Total preservation of information type
Structural_lability	all off-diagonals	Degree of structural reformatting
Phase_exchange	`xty + ytx`	Even/odd or cross-phase coupling
Phase_bias	`xty - ytx`	Directional phase asymmetry
Broadcast	`xtr + ytr + str`	Information becomes shared in the future
Fragmentation	`rtx + rty + stx + sty`	Distributed information becomes localized
Integration_formation	`rts + xts + yts`	Information becomes synergistically integrated
Integration_deployment	`str + stx + sty`	Synergy resolves into simpler forms
Net_integration_balance	formation - deployment	Net tendency toward integration or unpacking
Source_asymmetry	x-row minus y-row	Whether p1 or p2 dominates
Target_asymmetry	x-column minus y-column	Whether t1 or t2 dominates

---

8. Multiscale Extension Across

These metrics become especially informative when tracked across different embedding lags .

Instead of interpreting each metric at one lag only, consider each one as a **curve across** :

`Metric( )`

Examples: - `Total_persistence( )` - `Phase_exchange( )` - `Net_integration_balance( )` - `Broadcast( )` - `Fragmentation( )`

This allows you to characterize:

- preferred timescales,
- scale-specific integration,
- transitions between persistent and emergent regimes,
- temporal scale invariance or scale selectivity.

Useful summary descriptors across

For any metric curve: - **peak** : timescale where the metric is maximal - **center of mass across** : weighted characteristic timescale - **entropy across** : whether the metric is narrowband or broadband - **smoothness / rigidity across** : whether the structure changes gradually or abruptly - **correlation between metrics across** : e.g. whether persistence and integration co-occur or trade off

This may be more informative than a single scalar TII value.

---

9. Practical Interpretation Examples

High Total Persistence + Low Structural Lability

- stable temporal regime
- information stays in the same form
- simple or strongly regular dynamics

High Phase Exchange

- alternating or oscillatory organization
- strong phase coupling across the embedding

High Broadcast

- diverse past structures converge toward a common future structure
- temporal stabilization or global spreading

High Fragmentation

- global or integrated structure breaks into local channels
- temporal specialization or partial breakdown of coherence

High Integration Formation

- local/shared signals combine into irreducible temporal wholes
- emergence of temporal patterns

High Integration Deployment

- integrated patterns are being expressed or unpacked into simpler future forms
- top-down constraint or resolution of a whole into parts

Positive Net Integration Balance

- the system tends to build higher-order temporal structure

Negative Net Integration Balance

- the system tends to deploy, collapse, or resolve integrated structure

---

10. Final Recommendation

For temporal PhiID, the most meaningful derived metrics are usually not those borrowed directly from the two-system setting, but those that capture:

1. **persistence vs reformatting**
2. **exchange vs asymmetry**
3. **broadcast vs fragmentation**
4. **integration formation vs integration deployment**
5. **their variation across**

A good practical default set is:

```
Total_persistence
Structural_lability
Phase_exchange
Broadcast
Fragmentation
Integration_formation
Integration_deployment
Net_integration_balance
```

These metrics preserve the interpretability of the 16 atoms while giving a cleaner temporal vocabulary for single-process embeddings.